

EE708: Fundamentals of Data Science and Machine Intelligence

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Module 1 Foundational Principles of Data Science and Machine Intelligence

Course Organization

- Instructor: Rajesh M. Hegde (rhegde@iitk.ac.in)
url : <https://home.iitk.ac.in/~rhegde/>
- TAs:
 - Deepali Kushwaha (deepkush@iitk.ac.in)
 - Shubham Dwivedi (shubhamd21@iitk.ac.in)
 - Priyadarshini Dwivedi (priyadw@iitk.ac.in)
 - Four others
- Website: Refer to the Course on **Hello IITK**
- Lecture Slots: Mon (5.15 PM), Wed (5.15 PM) - 75 Minutes
- Lecture Venue: L-17 (Except when senate meetings – we will move to L-18)

Course Pre-requisite

- Statistics and probability
- Linear Algebra
- Numerical Methods like Gradient Descent (Differentiation and Integration)
- Data Representation
- Python Programming
- **An open mind to learn along with the Instructor**

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Course Content

- **Module 1:** General Introduction and Overview of data science and machine intelligence techniques
- **Module 2:** Statistical data analysis, visualization, and inference
- **Module 3:** Regression analysis and modeling: Linear, Non-Linear, Logistic
- **Module 4:** Clustering, Vector Quantization, Trees, Forests, Bagging, Boosting
- **Module 5:** Gaussian Mixture modeling
- **Module 6:** Artificial Neural Networks
- **Module 7:** Deep Neural Networks, Convolutional Neural Networks, Recurrent Neural Networks, and Autoencoders
- **Module 8:** Applications of machine intelligence in science and engineering

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Reference Books : Refer to Course Web @ Hello IITK

- Alpaydin, E., Introduction to Machine Learning. Cambridge, MA: MIT Press. ISBN: 9780262043793, Mar. 2020
- Duda and Hart, Pattern Classification, Wiley, 2001
- C. Bishop, Pattern Recognition and Machine Learning, Springer, 2006.
- Goodfellow, I.; Bengio, Y. & Courville, A., Deep Learning, MIT Press, 2016.
- S. Theodoridis and K. Koutroumbas, Pattern Recognition Second Edn, Elsevier, 2003
- B. Yegnanarayana, Artificial Neural Networks, PHI, 1999.
- Simon Haykin, Neural Networks and Learning Machines, Pearson, 1999.
- Joel Grus, Data Science from Scratch: First Principles with Python, O'Reilly Media, Inc 2019
- Andreas C. Müller, Sarah Guido, Introduction to machine learning with Python : a guide for data scientists, O'Reilly Media, Inc., 2016
- Andriy Burkov, The Hundred-Page Machine Learning Book, 2019
- Jake VanderPlas, Python Data Science Handbook: Essential Tools for Working with Data, O'Reilly Media, Inc 2016
- Wes McKinney, Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython, O'Reilly Media, Inc 2017
- Aurélien Géron, Hands-On Machine Learning with Scikit-Learn and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems, O'Reilly Media, Inc 2017
- Thomas Nield, Essential Math for Data Science, O'Reilly Media, Inc., 2022
- Peter Bruce and Andrew Bruce, Practical Statistics for Data Scientists, O'Reilly Media, Inc, 2017
- John D. Kelleher, Brian Mac Namee, Aoife D'Arcy, Fundamentals of Machine Learning for Predictive Data Analytics: Algorithms, Worked Examples, and Case Studies, MIT Press, 2015
- John Hearty, Advanced Machine Learning with Python, Packt Publishing, 2016
- Kevin Patrick Murphy, Probabilistic Machine Learning: Advanced Topics, MIT Press, 2023

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Grading Policy

- Assignment, Course Project material Uploads and Evaluations will be digital via Hello IITK**
- Mid Term Exam, End Sem Exam will be Physical Paper based exams**

Task	No. of Questions	% Contribution to Overall Grade
6 Assignments	Variable	15
Course Project	Report, Presentation and Demo	25
Mid Semester Exam	Variable	25
End Semester Exam	Variable	35

**No copying of assignments and In Exams.
Learn from the Course. You are definitely better than a copycat**

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Introduction

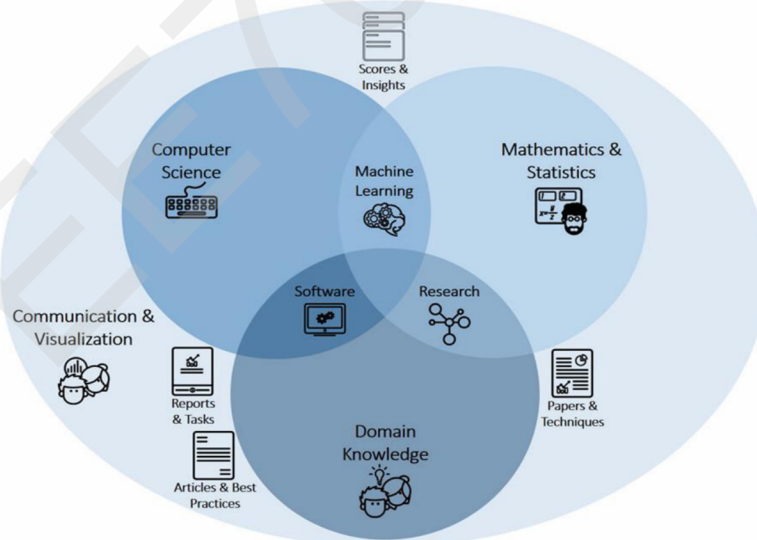
- **Data science** is not a single discipline.
- It integrates diverse expertise and skills to solve problems and optimize processes.
- The most important skills are **mathematics and statistics, computer science, and domain knowledge**.
- **Data scientists** use mathematics and statistics to analyze and model data from diverse fields, gaining insights or predicting future events.
- The diverse fields include – pure sciences, business, healthcare, space research, infrastructure, 5G/6G communication, signal processing, etc.
- **Mathematics and statistics** are essential for evaluating models, ensuring they fit the problem, and improving processes.

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Data Science : Expertise Areas



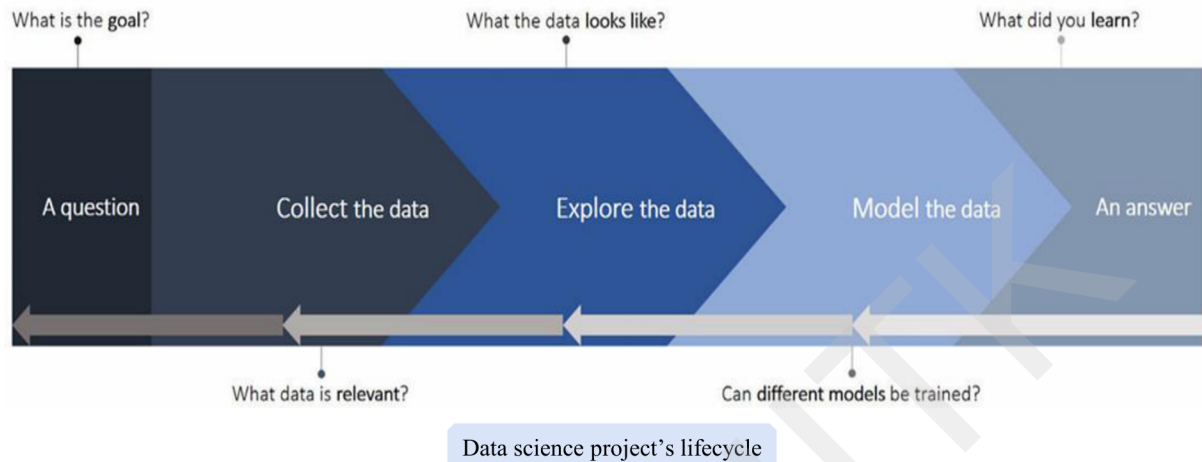
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Data Science Lifecycle

Steps in the analytical lifecycle, or process flow in Data Science



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Data Science Practical Applications

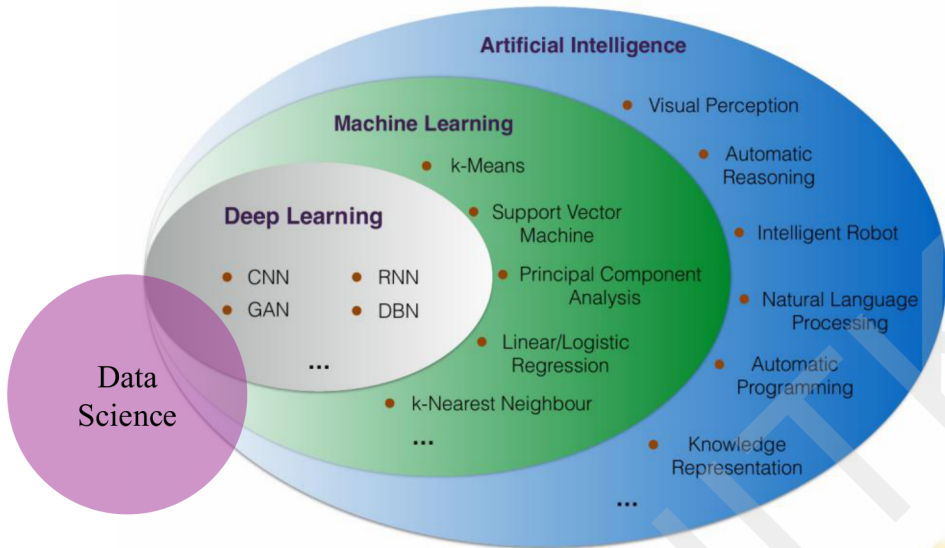


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AI, Machine Learning, Deep Learning, and Data Science



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Machine Learning

Machine learning is the science (and art) of programming computers so they can learn from data.

John McCarthy, 1956

AI: The science and engineering of making intelligent machines.

Arthur Samuel, 1959

ML: A field of study that gives computers the ability to learn without being explicitly programmed.

Mitchell, 1998

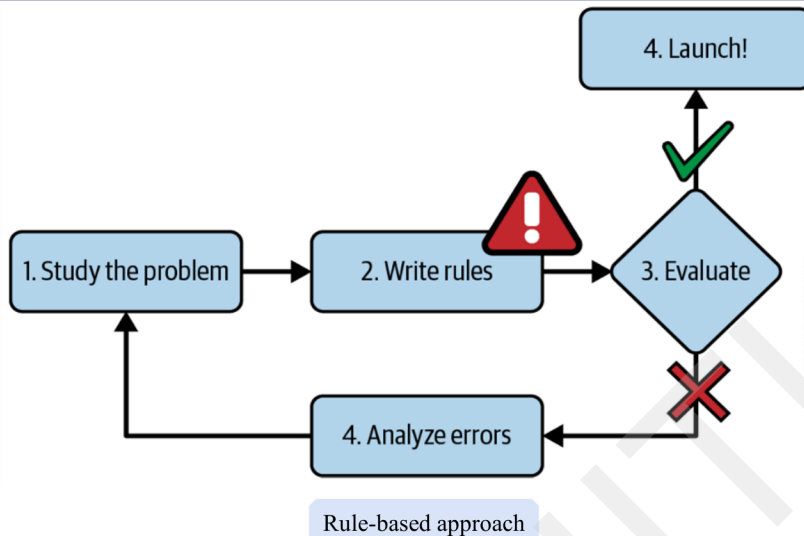
ML: A computer program is said to **learn from experience E**, with respect to some **task T**, and **some performance measure P**, if its **performance on T**, as measured by **P**, improves with experience **E**.

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Traditional Approach to Building/Deploying Applications

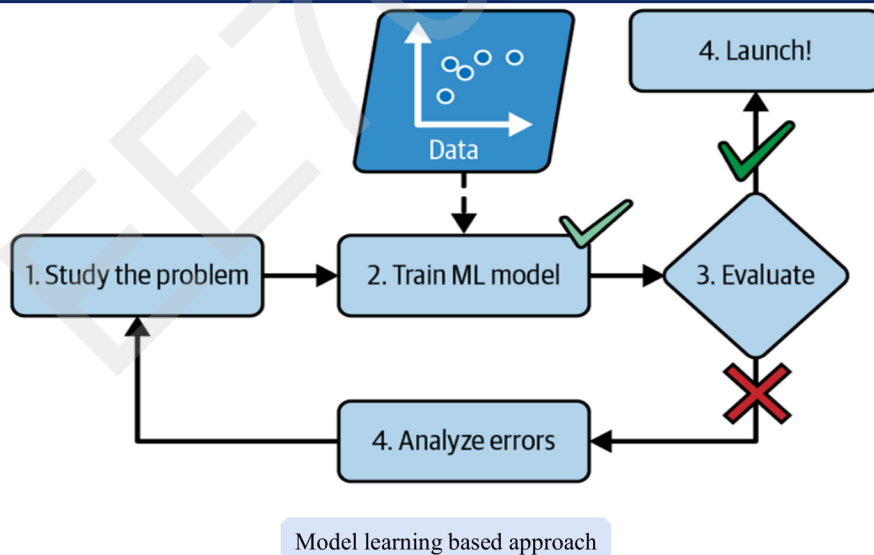


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Machine Learning Approach to Building/Deploying Applications



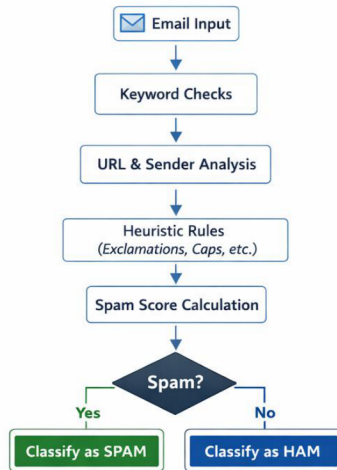
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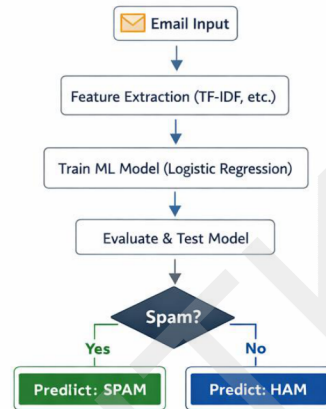
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Machine Learning Approach to Building/Deploying Applications

Rule-Based Spam Detection



Machine Learning-Based Spam Detection



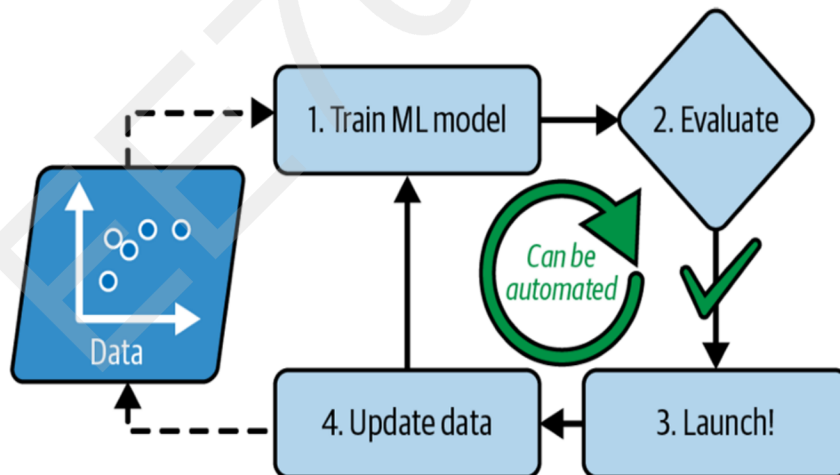
ChatGPT prompt: generate a flow diagram for rule based and ML based spam classification

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Machine Learning : Dynamic Adaptation to Solve Problems



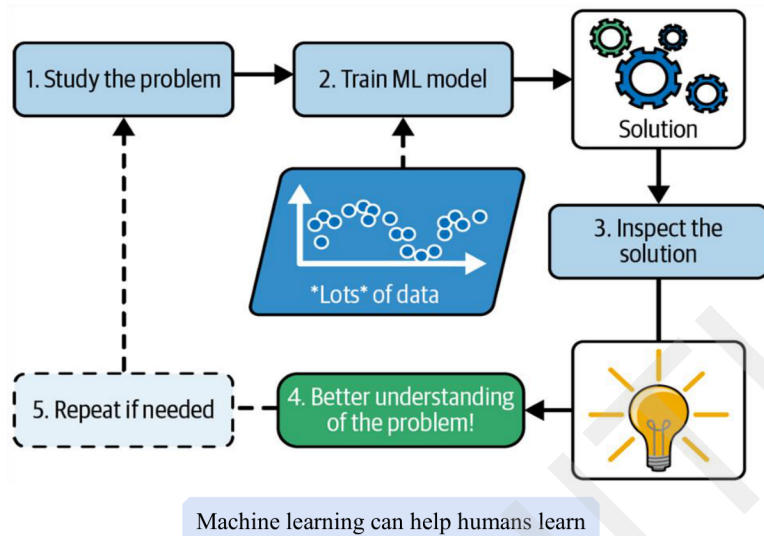
Automatically adapting to change

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Machine Learning : Gain New Perspectives of the Problem

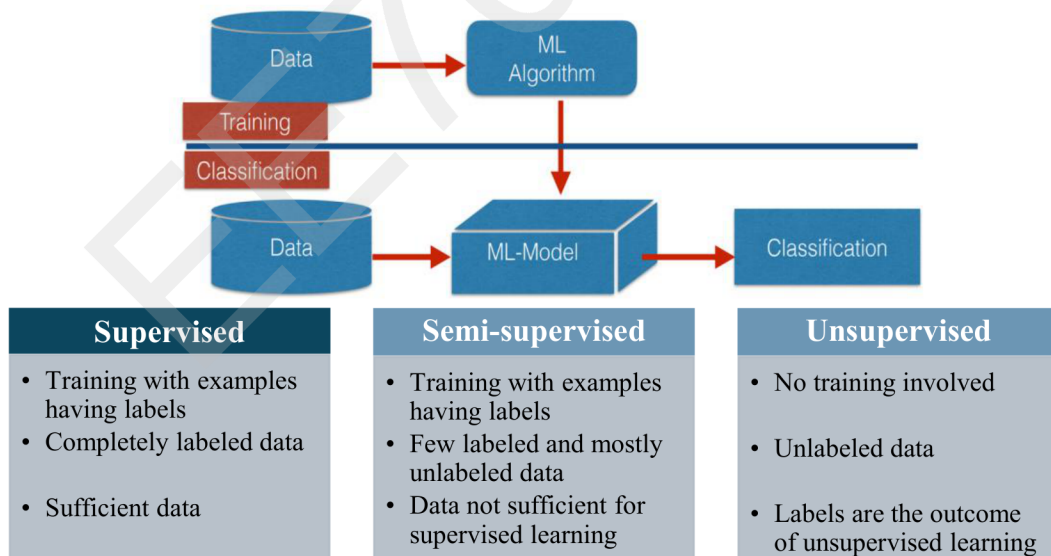


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Machine Learning : Overview, Classification/Types

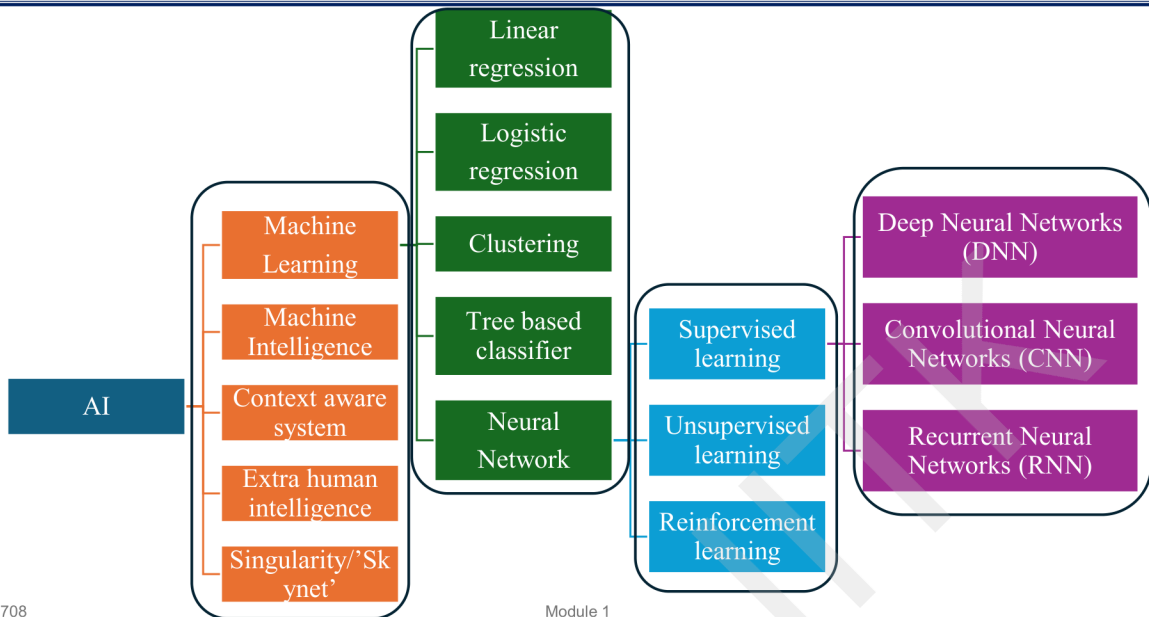


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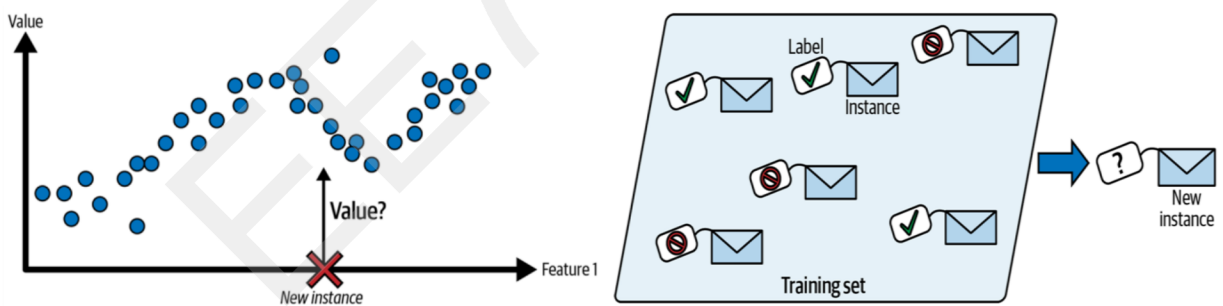
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Machine Learning Concepts and Techniques



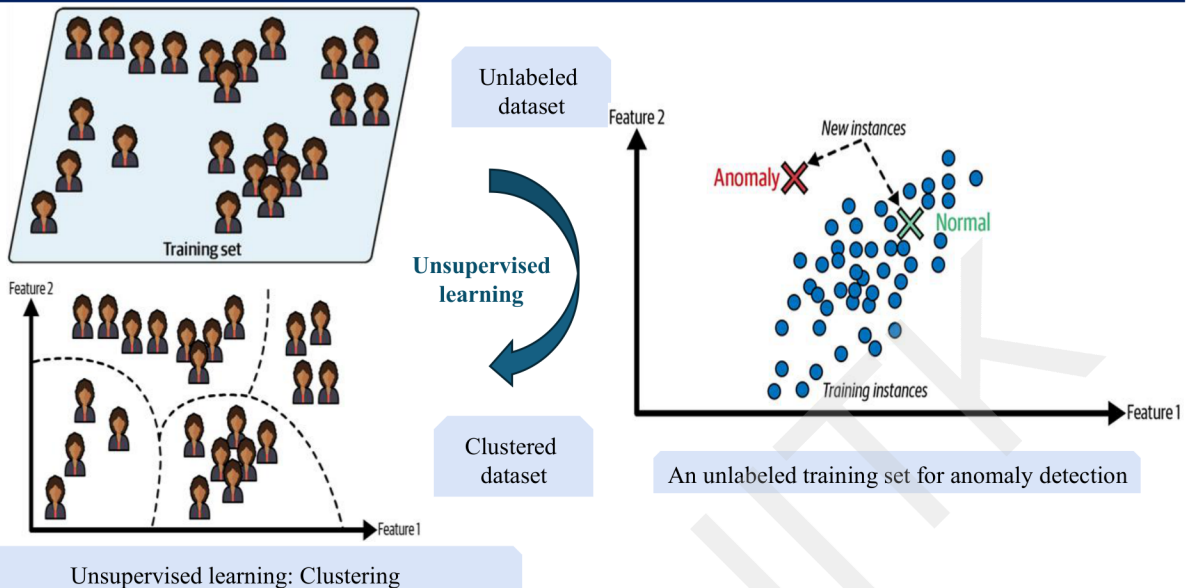
Supervised Learning



Regression problem: predict a value, given an input feature

A labeled training set for spam classification

Unsupervised Learning



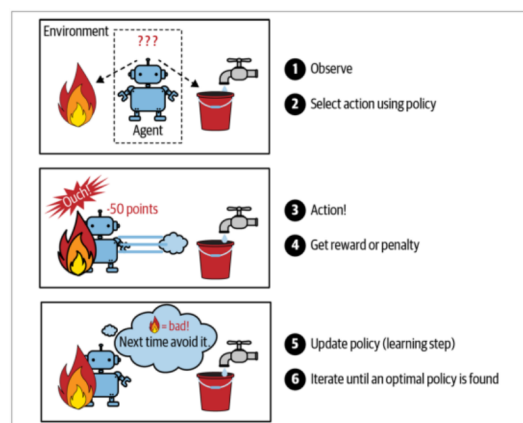
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Reinforcement Learning

- The learning system is referred to as an **agent**.
- Agent Capabilities:
 - Observe:** The agent can observe the environment.
 - Act:** It can select and perform actions within the environment.
 - Rewards:** It gets rewards/ penalties (negative rewards) for actions taken.
- The agent must learn by itself to determine the best strategy, called **policy**.
- A policy defines the action the agent should choose in any given situation.
- The policy aims to maximize the total reward over time.



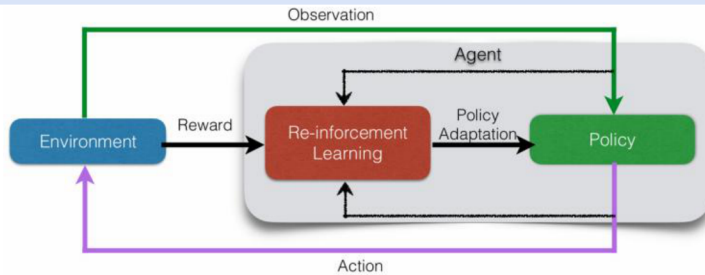
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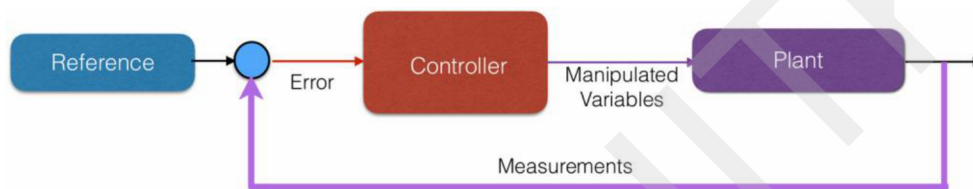
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Reinforcement Learning: Analogy from Control System

Reinforcement Learning



Control System



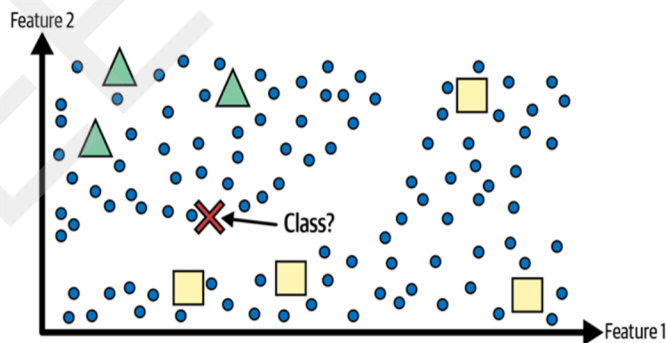
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Semi-supervised Learning

- Labeling data is usually time-consuming and costly.
- There are plenty of unlabeled instances and few labeled instances.
- Algorithms that can deal with data that's partially labeled is called semi-supervised learning.



Semi-supervised learning with two classes (triangles and squares). The unlabeled examples (circles) help classify a new instance (the cross) into the triangle class rather than the square class, even though it is closer to the labeled squares

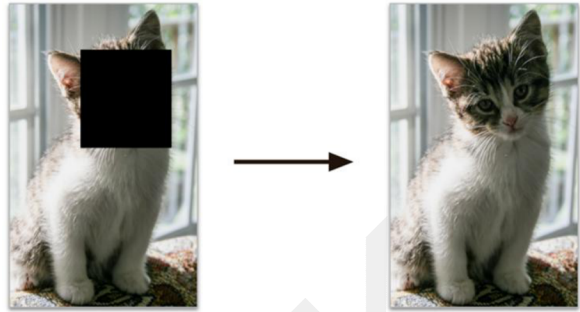
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Self-supervised Learning

- Generates a fully labeled dataset from a fully unlabeled one.
- After the whole dataset is labeled, any supervised learning algorithm can be used.
- Example: randomly mask small part of images and train model to recover original image.
- Use masked images as inputs and original images as labels.



Self-supervised learning example

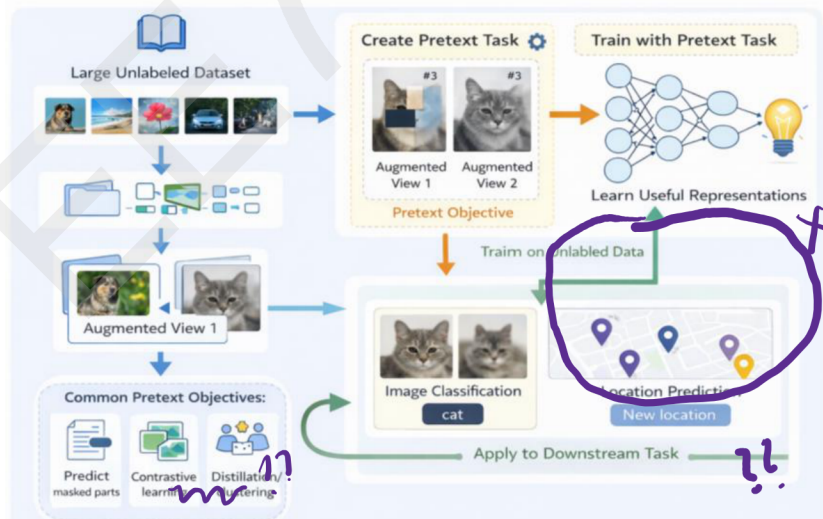
- The resulting model can repair damaged images or erase unwanted objects from pictures.
- Given a large dataset of unlabeled photos of pets – train an image-repairing model using self-supervised learning.
- Once the model is trained well, it can also distinguish pet species.
- Can be used for varying ML tasks.

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Self-supervised Learning

Self-Supervised Learning

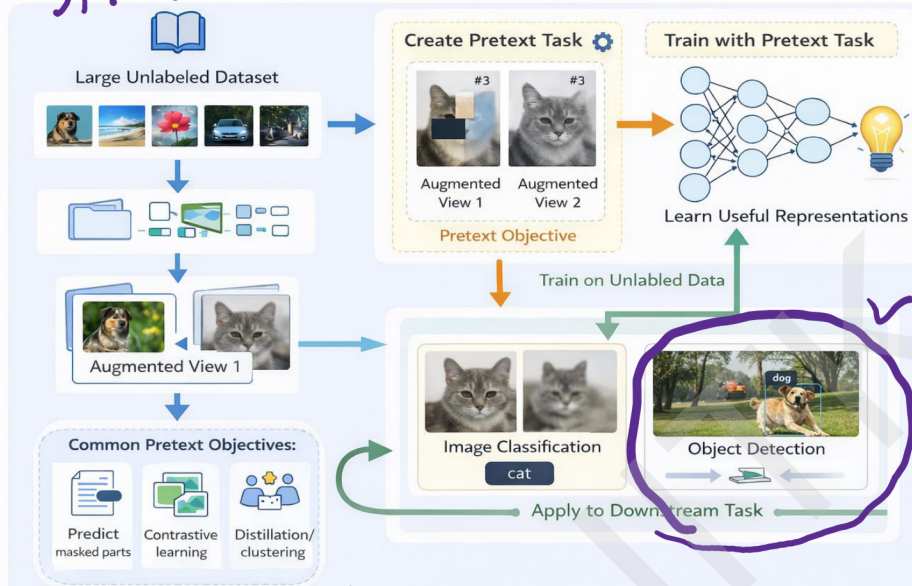


ChatGPT prompt: create a general figure for self supervised learning with a flow diagram

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On Questg.
Chatgpt corrects

Self-Supervised Learning

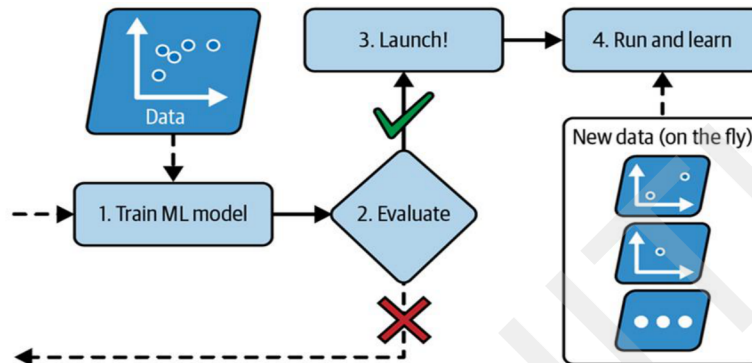


Learning Methods: Batch Learning

- **No Incremental Learning:** Batch learning systems require training on the entire dataset and cannot adapt incrementally.
- **Offline Training:** Training is done offline, consuming significant time and computational resources.
- **Static Application:** Once deployed, the model applies learned patterns without further updates.
- **Performance Decay:** Over time, the model's performance deteriorates due to changes in real-world data, a phenomenon called *model rot* or *data drift*.
- **Updating Process:** To update, the system must be retrained from scratch using the entire dataset (old and new), and the old model is replaced with the new one.

Learning Methods: Online Learning

- Learns incrementally by processing data instances sequentially or in mini-batches.
- Each learning step is fast and efficient, enabling real-time adaptation to new data.
- Suitable for scenarios with limited computing power, like training on mobile devices.
- Example : Detecting stock market trends.



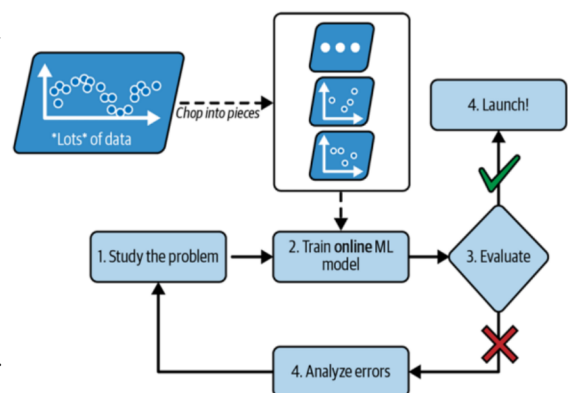
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Learning Methods: Online Learning to handle Huge Data Sets

- Training on massive datasets that exceed a machine's memory by processing data in smaller chunks (out-of-core learning).
- The system iteratively loads data, updates the model, and repeats until all data is processed.
- The *learning rate*, determines adaptation speed:
 - **High learning rate**: Fast adaptation but risks forgetting older data.
 - **Low learning rate**: Slower learning but better retention and resistance to noise or outliers.



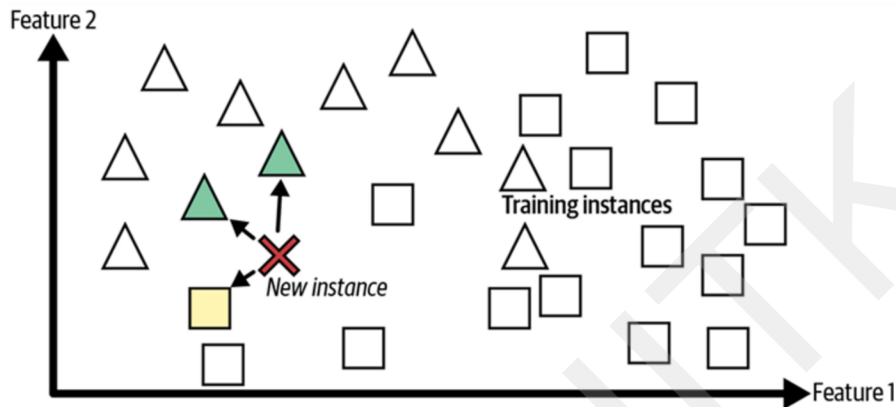
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Learning Methods: Instance based Learning

- Learns the examples by heart
- Generalizes to new cases by using a similarity measure to compare them to the learned examples (or a subset of them).



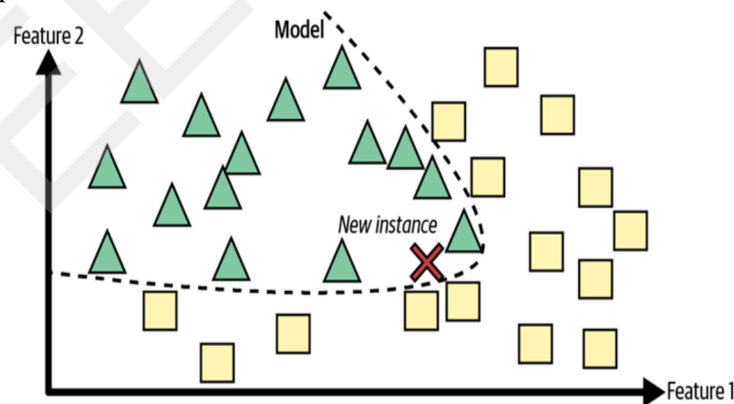
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Learning Methods: Model-based Learning

- Typical machine learning workflow
- Generalize from a set of examples: Build a model of these examples and then use that model to make predictions.



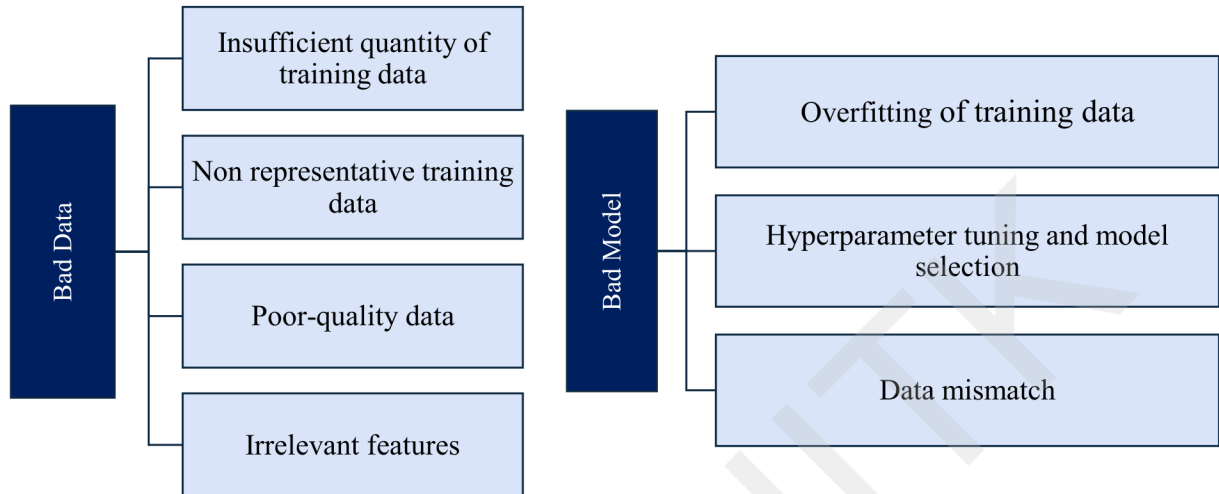
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Machine Learning: Challenges

- Two things that can go wrong are “Bad Data” and “Bad Model”.



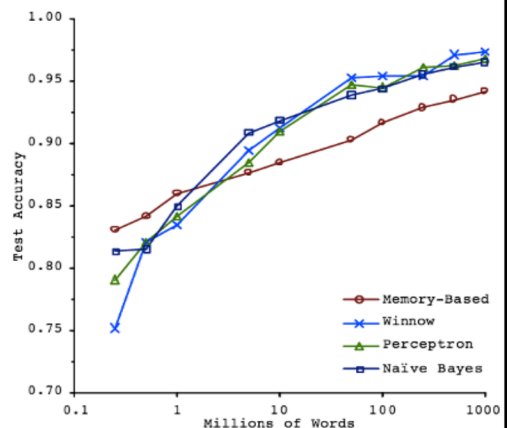
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Machine Learning: Dataset Vs Algorithms

- Michele Banko and Eric Brill (Microsoft Research) showed various machine learning algorithms, including simpler ones, achieved almost the same performance on the complex task of natural language disambiguation when given sufficient data.*
- As the authors put it, "... these results suggest that we may want to reconsider the trade-off between spending time and money on algorithm development versus spending it on corpus development".



"It's not who has the best algorithm that wins, it's who has the most data." - Andrew Ng

* Michele Banko and Eric Brill, "Scaling to very very large corpora for natural language disambiguation." In Proceedings of the 39th Annual Meeting on Association for Computational Linguistics (ACL '01). Association for Computational Linguistics, USA, 26–33, 2001

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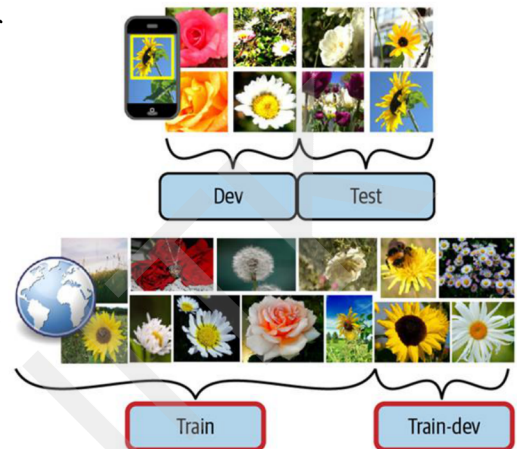
Machine Learning : Using the Dataset for Optimal Results

• Data mismatch

- In some cases, it's easy to get a large amount of data for training, but this data probably won't be perfectly representative of the data used in production.
- One solution is to hold out some of the training pictures in another set *train-dev set*.

• **Example:** Create a mobile app to take pictures of flowers and automatically determine their species.

- D1: Download millions of pictures of flowers on the web, but they won't be perfectly representative of the pictures that will actually capture using the app.
- D2: Let 1,000 representative pictures (i.e., actually taken with the app).

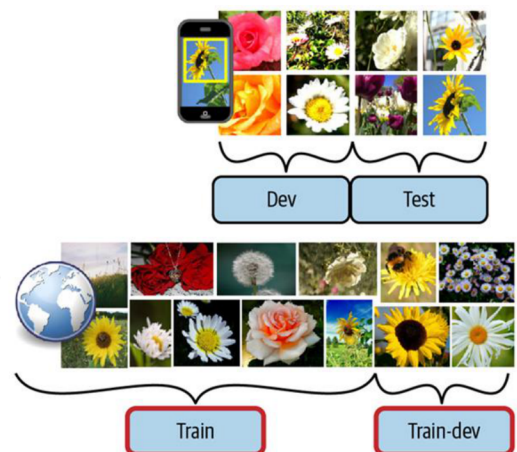


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Machine Learning : Using the Dataset for Optimal Results

- Shuffle D2 and put half in the validation/dev set and half in the test set (making sure that no duplicates or near-duplicates in both sets).
- Train the model with D1 and validate using D2 dev set
- Performance is disappointing implies
 - model has overfit the training set
 - mismatch between the web pictures and the mobile app pictures
- Solution : Hold out some of D1 data as **train-dev** set and validate the model on train-dev set.
 - Poor performance implies overfitting
 - Good performance implies need to further validate on D2 dev set
 - On obtaining poor performance it may imply data mismatch (preprocessing the D1 data)



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Machine Learning : No Free Lunch Theorem

- A model is a simplified representation of the data.
- Model selection depends on the assumptions about the data.
- Example: to choose a linear model, the assumption is “data is linear and the distance between the instances and the straight line is noise, which can be ignored.
- **In 1996 paper, David Wolpert demonstrated no assumption about the data, then there is no reason to prefer one model over any other. This is called the *No Free Lunch (NFL)* theorem.**
- For some datasets, the best model is a linear model, while for other datasets, it is a neural network.
- There is no model that is a priori guaranteed to work better (hence the name).
- The only way to know for sure which model is best is to evaluate them all.
- Since this is not possible, in practice some reasonable assumptions about the data are made and are evaluated only a few reasonable models.

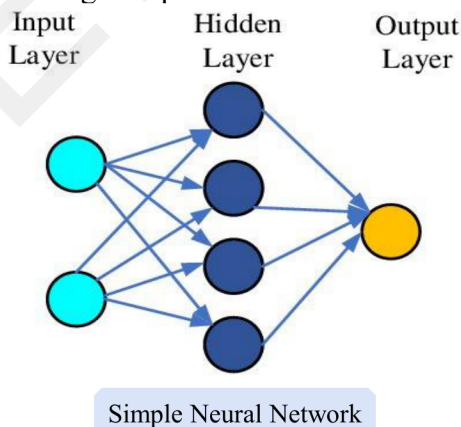
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Neural Networks : Overview

- A typical neural network is a group of algorithms, these algorithms models the data using neurons for machine learning.
- Neural networks are deep learning that powers the most human-like intelligence artificially.



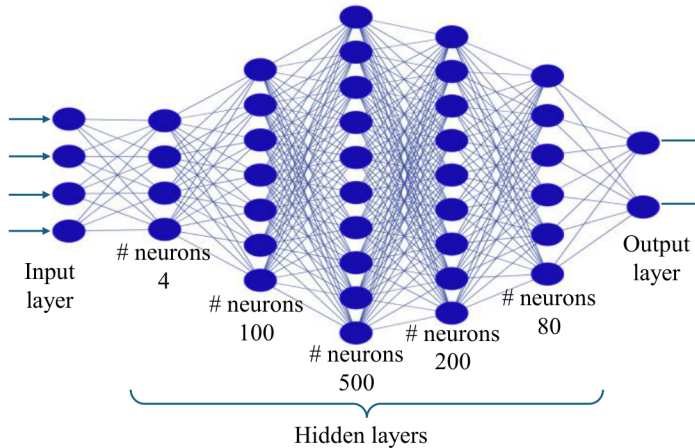
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Deep Neural Networks (DNN): Structure and Types

- DNN uses multiple layers between input and output layer to progressively extract higher level features from the raw input.
- Learning can be supervised, semi-supervised or unsupervised.



Multi-layer perceptron
neural networks

Convolutional neural
networks

Recurrent neural networks

Convolutional recurrent
neural networks

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Applications

- Classify products on a production line
- Detecting tumors in brain scans
- Classifying news articles
- Flagging offensive comments on discussion
- Summarizing long documents
- Creating a chatbot or a personal
- Forecasting a company's next year's revenue
- Making the app react to voice commands
- Detecting credit card fraud
- Segmenting clients and designing different marketing strategies
- Representing a complex, high-dimensional dataset in a clear and insightful diagram
- Recommending a product that a client may be interested in, based on past purchases
- Building an intelligent bot for a game

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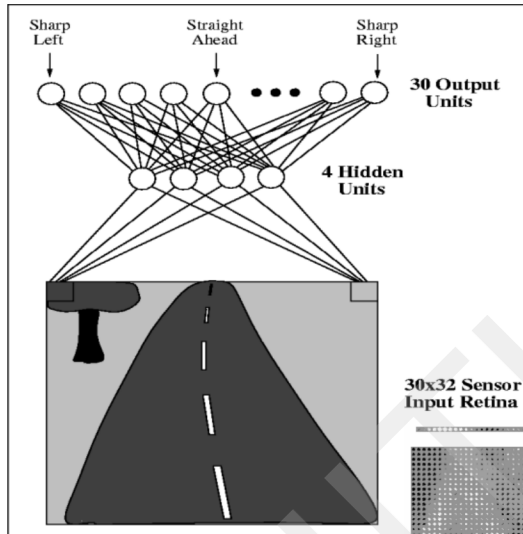
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Windup : Tasks too Hard to Program



ALVINN: steer autonomous vehicles by observing human driving capable of highway speeds up to 70 mph.



Architecture

- Input Layer: 30x32 "retina" receiving video input.
- Hidden Layer: **Five units using backpropagation**
- Output Layer: 30 units indicating steering direction.

Alvin Project Homepage, http://www.rl.cmu.edu/projects/project_160.html

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